Advanced probability

STAT6010/7610

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Course information

Contact, times, locations

Instructor: HOFERT Marius (`mhofert@hku.hk`)

Lectures: Mon 09:30–12:20, in MWC T7

Office hours: on appointment, via Zoom:
`https://hku.zoom.us/j/98597702554`

Questions: Best asked during the lecture breaks, after the lectures and during the office hours. Emails to the instructor will only be answered if they concern personal circumstances or emergency cases.

TA(s): YAO Gan (`ganyao@connect.hku.hk`; responsible for tutorials (starting in second week of classes) and grading assignments)

Representative: Any volunteer?

Course objectives

- Introduction to measure theory and probability
- Basic concepts in theoretical probability
- For students interested in research (in AS, STAT, probability)

- Mathematical proofs and the underlying ideas

References

It is recommended to study from the course material rather than additional sources. The course material is compiled from various sources, including:

- Achim Klenke, "Probability Theory: A comprehensive course", 2008;
- René Schilling, "Measures, integrals and martingales", 2006;
- Heinz Bauer, "Measure and Integration Theory", 2001;
- Gerald B. Folland, "Real Analysis: Modern Techniques and Their Applications", 1999;
- Rick Durrett, "Probability: Theory and Examples", edition 5, 2019;
- Sidney I. Resnick, "A Probability Path", 2014;
- Allan Gut, "Probability: A Graduate Course", 2005;
- David Williams, "Probability with Martingales", 1991;
- Patrick Billingsley, "Probability and Measure", 1995;
- Jean Jacod, Philip Protter, "Probability Essentials", 2003;

- Richard L. Wheeden, Antoni Zygmund, "Measure and Integral", 1977.

Teaching and assessment

- 2025-02-24: No lecture
- Course-relevant material is shared on Moodle (`https://moodle.hku.hk/course/view.php?id=122012`).
- **Assessment:**
  - 3 assignments (each 5%): to be handed in as Moodle Assignment (single file) before the tutorial on the due date (late hand-ins are marked as 0)
  - 1 midterm (25%): March 17, 2025, 09:30–10:20, MWC T7 (lecture thereafter)
  - 1 2 h final (60%): t.b.a. in course outline once available ($\approx$ mid term)
- **Absence from assessments:** See course outline. Avoid missing the final.
- See course outline for additional details, e.g. rules for regrading requests.

Absence from assessments

See course outline.

General advice

- Come to every class (even if you cannot follow much). Fill gaps in follow-up course work.
- Perform follow-up course work (after each lecture) to learn the material continuously throughout the term. For a proper learning effect and to prepare you for the exams, you should write by hand (definitions, main results, formulas, doing the examples and exercises again, etc.), so writing your own summary notes is advisable.
- Try every assignment question on your own first before collaborating with others. If you cannot solve them in a reasonable amount of time, discuss ideas with others. As in exams, the final write-up to be handed in must be your own.
- Try your best to avoid getting ill before your finals. It is highly recommended to participate in final exams, as supplementary finals are more difficult due to the longer preparation time, which would otherwise be unfair towards all other students.
- Regularly check your university email (especially before exams, lectures in bad weather).

FAQ

- *Is there a single book for this course we follow?*
  No (as all have their drawbacks and the lecture material is compiled from various sources).
- *Can I have more space on the slides?*
  In LaTeX:

```
\documentclass{article}
\usepackage{pdfpages}
\begin{document}
\includepdf[pages={1-}, scale=0.75, offset=0 180]{mySlides.pdf}% see https://mirror-hk.koddos.net/CTAN/macros/
      latex/contrib/pdfpages/pdfpages.pdf for the arguments; play with 'offset' to put slides in the location
      you want
\end{document}
```

- *What will the exam(s) look like?*
  Closed book. A mix of "Show...", "Calculate...", "Provide a definition of...", "Explain in words why...", etc. Important is to justify your answers (give reasons for your answers; providing an answer without derivation or without stating assumptions will only give a minor fraction of the marks).
- *How can I get more practice?*
  You can...

- rework the course slides (definitions, statements, examples);
- change the distributions in the examples and redo them;
- redo the assignment and tutorial problems (also with different numbers);
- team up with a colleague and pass each other (modified) questions; and
- google the topics you struggle with to find more exercises.

- *Do I need to be able to replicate all the proofs?*
  Proofs are an essential part of mathematical learning and are covered to explain why statements are correct, so they help us learn and understand. Important arguments and rough ideas (for longer proofs) may appear or be asked for, but longer proofs do not need to be replicated precisely.

Overview

1 Introduction

2 Measure theory

3 Measurable mappings

4 Ordinary conditional probability, independence and dependence

5 Integration and expectation

6 Modes of convergence

7 Characteristic functions

8 Conditional expectation

9 Martingales

1 Introduction

1.1 History

1.2 Basics of set theory

1.1 History

- Intuitively, we understand the concept of probability (a number in $[0, 1]$ reflecting the chance that a random event happens) and that is important for modelling future events (gambling, weather, life insurance, success of a new drug).
- Early references include:
  - Fair value of an insurance contract, e.g. against crop failure (Code of Hammurabi from Babylon; legal text from 1754 BCE):

    > "A farmer who has a mortgage on his property is required to make annual interest payments in the form of grain. However, in the event of a crop failure, this farmer has the right not to pay anything, and the creditor has no alternative but to forgive the interest due."

    This essentially describes a put option (right but not the obligation of the holder [here: farmer] to sell [here: not to pay interest in the form of crop] the underlying asset [here: crop] at a specific price).
  - Gerolamo Cardano (1501–1576, "Book on Games of Chance" (on gambling; written ≈ 1564; considered throwing dice to understand basic probabilities; considered the ratio of favourable to unfavourable outcomes as probabilities)

- Galileo Galilei (1564–1642) observed that some numbers in $\{2, \ldots, 12\}$ appear more often as sum when throwing two dice since there are more ways to create them.
- Blaise Pascal (1623–1662) and Pierre de Fermat (1607–1665) exchanged letters to solve gambling "paradoxes", commonly viewed as the birth of probability theory.
  - Antoine Gombaud (as self-styled Chevalier de Méré) considered two games (Chevalier de Méré paradox):

    1) Roll a fair die 4x and note whether at least one 6 occurs.

    2) Roll two fair dice 24x and note whether at least one double 6 occurs.

    He believed the chance of the two games to be the same (they are $1-(1-1/6)^4 \approx 0.5177$ and $1-(1-1/36)^{24} \approx 0.4914$, respectively).
  - Pascal and Fermat spotted that Gombaud believed that the probability of success in n throws is n times that of a single throw; e.g. in Game 1 Gombaud thought that since the probability of success in one throw is $1/6$, the probability of rolling at least one 6 in four throws is $4/6 = 2/3$.
- Christiaan Huygens (1629–1695) wrote a book about the ideas of Pascal

and Fermat.

- Jacob Bernoulli (1655–1705) pointed out the necessity to develop a theory to answer interesting probability problems he proposed in 1685.
- More and more mathematicians worked on probability problems: Abraham de Moivre (1667–1754), Daniel Bernoulli (1700–1782), Leonhard Euler (1707–1783), Carl Friedrich Gauss (1777–1855), Pierre-Simon Laplace (1749–1827).
- Andrey Nikolaevich Kolmogorov (1903–1987) saw the usefulness of measure theory in properly establishing a modern theory of probability in his book "Grundbegriffe der Wahrscheinlichkeitsrechnung" in 1933. Measure theory is indispensable for studying probability theory.
- Paul Lévy (1886–1971) worked on stochastic processes, characteristic functions and limit theorems.

- Today we think of probability as a mathematical theory for modelling random events.
- Random events are described through sets. We thus first need to learn about basic set theory and families of sets with certain properties. We can then define probabilities on such families of sets in a consistent (non-contradicting) way.

1.2 Basics of set theory

- Under a *collection* we understand several objects (*elements*) grouped together.
- Initially, a "set" was believed to be an arbitrary collection of objects, its elements.
- *Russell's paradox* (1901) showed that every set theory that allows *unrestricted comprehension* (i.e. that for any well-defined property we can construct a "set" containing the elements with that property) leads to contradictions since such a general "set" can be an element of itself, e.g. $\{A : |A| \geq 1\}$ is a "set" which contains itself (e.g. $\{e, \pi\} \in \mathcal{A} := \{A \subseteq \mathbb{R} : |A| \geq 1\} \Rightarrow \mathcal{A} \in \mathcal{A}$).
- Russell considered the "set" $R := \{A : A \notin A\}$ of all "sets" that are not elements of themselves. If $R \notin R$, then $R \in R$. And if $R \in R$, then $R \notin R$. So $R \in R \Leftrightarrow R \notin R$ ↯. A more colloquial expression of this problem is: "This statement is false."

  Barber paradox: Suppose a barber shaves precisely those men who do not shave themselves, does the barber shave himself?

  - If yes, then he shaves himself, so the barber (he) does not shave himself ↯.
  - If no, then he doesn't shave himself, so the barber (he) shaves himself ↯.

- Ernst Zermelo (1871–1953), with later additions by Abraham Fraenkel (1891–1965), developed a system of axioms in order to formulate a set theory free of paradoxes, the Zermelo–Fraenkel (ZF) set theory; it does not allow for the existence of a universal set (a set containing all sets; this often leads to problems) nor for unrestricted comprehension (so it also avoids Russell's paradox). With the *axiom of choice* (i.e. for any collection of non-empty sets $\mathcal{A}$ there is a *choice function* f such that $f(A) \in A$ for all $A \in \mathcal{A}$; or: the Cartesian product of a collection of non-empty sets is non-empty), ZF is abbreviated ZFC. We assume to work with ZFC.
- A *set* is a collection of objects (*elements*) that satisfy the ZFC axioms.
- Let $\Omega \neq \emptyset$. A is a *subset* of Ω ($A \subseteq \Omega$) or Ω is a *superset* of A ($\Omega \supseteq A$), if $\omega \in A \Rightarrow \omega \in \Omega \; \forall \omega \in A$. And $A = \Omega$ if $A \subseteq \Omega$ and $\Omega \subseteq A$.
- $\mathcal{P}(\Omega) := \{A : A \subseteq \Omega\}$ is the *power set* of Ω, i.e. the set of all subsets of Ω (including $\emptyset, \Omega$). If $|\Omega| < \infty$, one can show by induction that $|\mathcal{P}(\Omega)| = 2^{|\Omega|}$.
- For $I \subseteq \mathbb{R}$, a *family* $\{A_i\}_{i \in I} \subseteq \mathcal{P}(\Omega)$ of sets is a collection of subsets of a set Ω. In contrast to a set of sets, a family of sets can contain repeated copies of its elements.

- Basic set operations:

Complementation: If $A \subseteq \Omega$, then $A^c := \{\omega \in \Omega : \omega \notin A\}$.
We have $(A^c)^c = A$, $\emptyset^c = \Omega$, $\Omega^c = \emptyset$.

Intersection: If $A_i \subseteq \Omega$, $i \in I$, then

$$\bigcap_{i \in I} A_i := \{\omega \in \Omega : \omega \in A_i \ \forall\, i \in I\}.$$

The family $\{A_i\}_{i \in I}$ is *pairwise disjoint* (or *mutually exclusive*) if $A_{i_1} \cap A_{i_2} = \emptyset \ \forall\, i_1, i_2 \in I : i_1 \neq i_2$. Intersections satisfy $A \cap B = B \cap A$ (*commutativity*) and $(A \cap B) \cap C = A \cap (B \cap C)$ (*associativity*).

Union: If $A_i \subseteq \Omega$, $i \in I$, then

$$\bigcup_{i \in I} A_i := \{\omega \in \Omega : \exists\, i \in I : \omega \in A_i\}.$$

If the family $\{A_i\}_{i \in I}$ is pairwise disjoint, one often writes $\biguplus_{i \in I} A_i$. Unions satisfy $A \cup B = B \cup A$ (*commutativity*) and $(A \cup B) \cup C = A \cup (B \cup C)$ (*associativity*).

Set difference: If $A, B \subseteq \Omega$, then $A \backslash B := A \cap B^c$. We thus have $A^c = \Omega \backslash A$.

- Identities involving more than one set operation:

  Distributivity: $A \cap \bigcup_{i \in I} A_i = \bigcup_{i \in I} (A \cap A_i)$ and $A \cup \bigcap_{i \in I} A_i = \bigcap_{i \in I} (A \cup A_i)$

  De Morgan's laws: $(\bigcup_{i \in I} A_i)^c = \bigcap_{i \in I} A_i^c$ and $(\bigcap_{i \in I} A_i)^c = \bigcup_{i \in I} A_i^c$

- Limits of sequences of sets:

  Infimum: $\inf_{k \geq n} A_k := \bigcap_{k \geq n} A_k$

  Supremum: $\sup_{k \geq n} A_k := \bigcup_{k \geq n} A_k$

  Limit inferior: $\liminf\limits_{n \to \infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{k \geq n} A_k \underset{\text{def.}}{=} \sup_{n \geq 1} \inf_{k \geq n} A_k$

  Limit superior: $\limsup\limits_{n \to \infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k \underset{\text{def.}}{=} \inf_{n \geq 1} \sup_{k \geq n} A_k$

  Limit: If $\liminf\limits_{n \to \infty} A_n = \limsup\limits_{n \to \infty} A_n =: A$, A is the *limit* of A_n for $n \to \infty$ and we write $A = \lim\limits_{n \to \infty} A_n$ or $A_n \underset{n \to \infty}{\to} A$.

- Monotone sequences of sets:

  $\{A_n\}_{n \in \mathbb{N}}$ is *increasing* ($A_n \nearrow$) if $A_1 \subseteq A_2 \subseteq \ldots$.

  $\{A_n\}_{n \in \mathbb{N}}$ is *decreasing* ($A_n \searrow$) if $A_1 \supseteq A_2 \supseteq \ldots$.

Lemma 1.1 (Interpretation limit inferior, limit superior of sets)

1) $\liminf\limits_{n\to\infty} A_n = \{\omega \in \Omega : \omega \in A_n \text{ for all but finitely many } n\} =: \{\omega \in A_n \text{ abfm}\}$
2) $\limsup\limits_{n\to\infty} A_n = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many } n\} =: \{\omega \in A_n \text{ io}\}$

Proof.

1) We have

$$\begin{aligned}\liminf_{n\to\infty} A_n &\underset{\text{def.}}{=} \bigcup_{n=1}^{\infty} \bigcap_{k\geq n} A_k \\ &= \{\omega \in \Omega : \exists\, n \in \mathbb{N} : \omega \in A_k \ \forall\, k \geq n\} \\ &= \{\omega \in \Omega : \omega \in A_n \text{ for all but finitely many } n\}.\end{aligned}$$

2) We have

$$\begin{aligned}\limsup_{n\to\infty} A_n &\underset{\text{def.}}{=} \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k \\ &= \{\omega \in \Omega : \forall\, n \in \mathbb{N}\ \exists\, k \geq n : \omega \in A_k\} \\ &= \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many } n\}.\end{aligned}$$

□

Lemma 1.2 (Properties of limit inferior, limit superior of sets)

We have

1) $\liminf\limits_{n\to\infty} A_n \subseteq \limsup\limits_{n\to\infty} A_n$.
2) $(\liminf\limits_{n\to\infty} A_n)^c = \limsup\limits_{n\to\infty} A_n^c$.

Proof.

1) $\liminf\limits_{n\to\infty} A_n = \{\omega \in A_n \text{ abfm}\} \subseteq \{\omega \in A_n \text{ io}\} = \limsup\limits_{n\to\infty} A_n$.
2) We have $(\liminf\limits_{n\to\infty} A_n)^c \underset{\text{def.}}{=} (\bigcup_{n=1}^\infty \bigcap_{k\geq n} A_k)^c \underset{\text{De Morgan}}{=} \bigcap_{n=1}^\infty \big(\bigcap_{k\geq n} A_k\big)^c \underset{\text{De Morgan}}{=} \bigcap_{n=1}^\infty \big(\bigcup_{k\geq n} A_k^c\big) \underset{\text{def.}}{=} \limsup\limits_{n\to\infty} A_n^c$. $\square$

Lemma 1.3 (Monotone sequences of sets)

1) If $A_n \nearrow$, then $\lim\limits_{n\to\infty} A_n$ exists and $\lim\limits_{n\to\infty} A_n = \bigcup_{k=1}^\infty A_k$. Similarly, if $A_n \searrow$, then $\lim\limits_{n\to\infty} A_n$ exists and $\lim\limits_{n\to\infty} A_n = \bigcap_{k=1}^\infty A_k$.
2) For all $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{P}(\Omega)$, $\liminf\limits_{n\to\infty} A_n = \lim\limits_{n\to\infty} (\inf_{k\geq n} A_k)$ and $\limsup\limits_{n\to\infty} A_n = \lim\limits_{n\to\infty} (\sup_{k\geq n} A_k)$.

Proof.

1) $\limsup\limits_{n\to\infty} A_n \underset{\text{def.}}{=} \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \underset{n\,\equiv\,1}{\subseteq} \bigcup_{k=1}^{\infty} A_k \underset{A_n\,\nearrow}{=} \bigcup_{k=1}^{\infty} \bigcap_{i=k}^{\infty} A_i \underset{\text{def.}}{=} \liminf\limits_{n\to\infty} A_n \underset{\text{L. 1.21)}}{\subseteq} \limsup\limits_{n\to\infty} A_n$, so that $\lim\limits_{n\to\infty} A_n$ exists and $\lim\limits_{n\to\infty} A_n = \bigcup_{k=1}^{\infty} A_k$. Similarly for $A_n \searrow$.

2) For $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{P}(\Omega)$, $\liminf\limits_{n\to\infty} A_n \underset{\text{def.}}{=} \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \underset{\text{def.}}{=} \bigcup_{n=1}^{\infty} \inf_{k\geq n} A_k \underset{1)}{\overset{\inf_{k\geq n} A_k \,\nearrow}{=}} \lim\limits_{n\to\infty} (\inf_{k\geq n} A_k)$. Similarly for $\sup_{k\geq n} A_k \searrow$. $\square$

- Equivalence relations: An *equivalence relation* is a binary relation $\sim$ on a set A (a set of ordered pairs from A) that is reflexive ($a \sim a$), symmetric ($a \sim b \Leftrightarrow b \sim a$) and transitive ($a \sim b$, $b \sim c \Rightarrow a \sim c$).
- The *equivalence class* of $a \in A$ under $\sim$ is $[a] := \{x \in A : x \sim a\}$. Example: "$=$" (is equal to), e.g. $1/2 = 2/4$, and both belong to the same equivalence class of cancelled fractions on $\mathbb{Q}$.
- Indicator functions: The *indicator function* of $A \subseteq \Omega$ is $\mathbb{1}_A(\omega) := \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$
  We have:
  1) $\mathbb{1}_{A^c} = 1 - \mathbb{1}_A$

2) $\mathbb{1}_A \leq \mathbb{1}_B$ (i.e. $\mathbb{1}_A(\omega) \leq \mathbb{1}_B(\omega)$ for all $\omega \in \Omega$) iff $A \subseteq B$

3) Indicator functions belong to the most useful functions there are. They can be used to count quantities, e.g.:

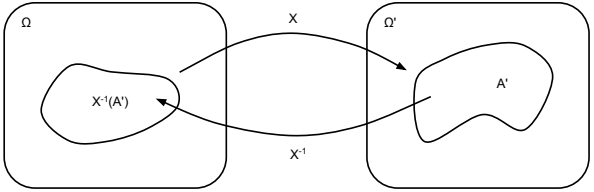
$$\limsup_{n\to\infty} A_n = \{\omega \in A_n \text{ io}\} = \left\{\omega \in \Omega : \sum_{n=1}^{\infty} \mathbb{1}_{A_n}(\omega) = \infty\right\},$$

$$\liminf_{n\to\infty} A_n = \{\omega \in A_n \text{ abfm}\} = \left\{\omega \in \Omega : \sum_{n=1}^{\infty} \mathbb{1}_{A_n^c}(\omega) < \infty\right\}.$$

- Preimages: The *preimage* of a map $X : \Omega \to \Omega'$ between two sets Ω, Ω' is

$$X^{-1}(A') = \{\omega \in \Omega : X(\omega) \in A'\}.$$

Sketch:



Preimages exist even if X is not injective (e.g. for $X(\omega) = \omega^2$, $X^{-1}([1,4]) = [-2,-1] \uplus [1,2]$).

- Preimages are closed with respect to (wrt) the following operations:

  Complementation: If $A' \subseteq \Omega'$, then $(X^{-1}(A'))^{c_\Omega} = X^{-1}(A'^{c_{\Omega'}})$.

  Proof. $\omega \in (X^{-1}(A'))^{c_\Omega} \Leftrightarrow \omega \notin X^{-1}(A') \Leftrightarrow X(\omega) \notin A' \Leftrightarrow X(\omega) \in A'^{c_{\Omega'}} \Leftrightarrow \omega \in X^{-1}(A'^{c_{\Omega'}})$. $\square$

  Union: If $\{A'_i\}_{i\in I} \subseteq \mathcal{P}(\Omega')$, then $X^{-1}(\bigcup_{i\in I} A'_i) = \bigcup_{i\in I} X^{-1}(A'_i)$.

  Proof. $\omega \in X^{-1}(\bigcup_{i\in I} A'_i) \Leftrightarrow X(\omega) \in \bigcup_{i\in I} A'_i \Leftrightarrow \exists\, i \in I : X(\omega) \in A'_i \Leftrightarrow \exists\, i \in I : \omega \in X^{-1}(A'_i) \Leftrightarrow \omega \in \bigcup_{i\in I} X^{-1}(A'_i)$. $\square$

  Intersection: If $\{A'_i\}_{i\in I} \subseteq \mathcal{P}(\Omega')$, then $X^{-1}(\bigcap_{i\in I} A'_i) = \bigcap_{i\in I} X^{-1}(A'_i)$.

  Proof. $\omega \in X^{-1}(\bigcap_{i\in I} A'_i) \Leftrightarrow X(\omega) \in \bigcap_{i\in I} A'_i \Leftrightarrow X(\omega) \in A'_i \;\forall\, i \in I \Leftrightarrow \omega \in X^{-1}(A'_i) \;\forall\, i \in I \Leftrightarrow \omega \in \bigcap_{i\in I} X^{-1}(A'_i)$. $\square$

  Monotonicity: If $\mathcal{A}', \mathcal{B}' \subseteq \mathcal{P}(\Omega')$, $\mathcal{A}' \subseteq \mathcal{B}'$ (i.e. $A' \in \mathcal{A}' \Rightarrow A' \in \mathcal{B}'$), then $X^{-1}(\mathcal{A}') := \{X^{-1}(A') : A' \in \mathcal{A}'\} \subseteq X^{-1}(\mathcal{B}')$.

  Proof. $A \in X^{-1}(\mathcal{A}') \Rightarrow A = X^{-1}(A')$ for some $A' \in \mathcal{A}'$ $\underset{\mathcal{A}' \subseteq \mathcal{B}'}{\Rightarrow} A = X^{-1}(A')$ for some $A' \in \mathrm{B}' \Rightarrow A \in X^{-1}(\mathcal{B}')$. $\square$

2 Measure theory

2.1 Non-measureable sets

2.2 Systems of sets

2.3 Measures

2.4 Probability measures

2.5 Null sets

2.6 Construction of measures

2.7 Borel measures on $\mathbb{R}$

2.8 Borel measures on $\mathbb{R}^d$, $d \geq 2$

- **Question:** A fundamental question in measure theory is to measure the volume (or size; or, later, probability) $\lambda(A) \geq 0$ of a set $A \subseteq \mathbb{R}^d (= \Omega)$, $d \geq 1$. Consider $d = 1$ (later also $d > 1$). How can this be done?
- A reasonable such $\lambda : \mathcal{F} \to [0, \infty]$ (some $\mathcal{F}$; λ for now informally called *measure*) should
  1) assign to an interval its length: $\lambda((a, b]) = b - a \; \forall\, a, b \in \mathbb{R} : a \leq b$;
  2) be invariant under translations, rotations and reflections: $\forall\, A, B \subseteq \mathbb{R}$ congruent, $\lambda(A) = \lambda(B)$;
  3) be *σ-additive*: If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(\mathbb{R})$, $A_i \cap A_j = \emptyset \; \forall\, i \neq j$, then $\lambda(\biguplus_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \lambda(A_i)$. At the moment it is not so clear why additivity (without the "$\sigma-$" part) is not sufficient to consider, T. 2.2 will address that.
- Note that λ must be *monotone* since $\lambda(B) = \lambda(A \uplus (B \backslash A)) \underset{3)}{=} \lambda(A) + \lambda(B \backslash A) \geq \lambda(A) \; \forall\, A \subseteq B$, another reasonable property.

2.1 Non-measureable sets

Question: Can we simply take $\mathcal{F} = \mathcal{P}(\mathbb{R}) = \{A : A \subseteq \mathbb{R}\}$ as domain? No!

Theorem 2.1 (Vitali's theorem)
There is no λ defined on $\mathcal{P}(\mathbb{R})$ which satisfies 1)–3).

Proof. We construct a set V, such that, when changed according to 1)–3), we obtain a contradiction.

- Consider $[0,1]$. Then $x \sim y :\Leftrightarrow x - y \in \mathbb{Q}$ defines an equivalence relation (reflexive, symmetric and transitive) on $[0,1]$, with equivalence classes $[x] := \{y \in [0,1] : y \sim x\}$, $x \in [0,1]$.
- The distinct equivalence classes of "$\sim$" partition $[0,1]$.
- The *Vitali set* contains precisely one element of each distinct equivalence class of "$\sim$", i.e.

$$V = \{v \in [0,1] : \ \forall\, x \in [0,1] \ \ \exists!\, v \sim x\};$$

  the construction requires the axiom of choice (we assumed to have in ZFC).
- Let $\{q_k\}_{k\in\mathbb{N}}$ be a unique enumeration of $\mathbb{Q} \cap [-1,1]$. This can be constructed with *Cantor's first diagonal argument*, here for $\mathbb{Q}$ (skip those numbers already covered to get a unique enumeration):

$$
\begin{array}{ccccccccccc}
0 \to \frac{1}{1} & \to & \pm\frac{1}{2} & & \pm\frac{1}{3} & \to & \pm\frac{1}{4} & & \pm\frac{1}{5} & \to & \pm\frac{1}{6} & \cdots \\
 & \swarrow & & \nearrow & & \swarrow & & \nearrow & & \swarrow & & \\
\pm\frac{2}{1} & & \pm\frac{2}{2} & & \pm\frac{2}{3} & & \pm\frac{2}{4} & & \pm\frac{2}{5} & & \pm\frac{2}{6} & \cdots \\
\downarrow & \nearrow & & \swarrow & & \nearrow & & \swarrow & & & & \\
\pm\frac{3}{1} & & \pm\frac{3}{2} & & \pm\frac{3}{3} & & \pm\frac{3}{4} & & \pm\frac{3}{5} & & \pm\frac{3}{6} & \cdots \\
 & \swarrow & & \nearrow & & \swarrow & & & & & & \\
\pm\frac{4}{1} & & \pm\frac{4}{2} & & \pm\frac{4}{3} & & \pm\frac{4}{4} & & \pm\frac{4}{5} & & \pm\frac{4}{6} & \cdots \\
\downarrow & \nearrow & & \swarrow & & & & & & & & \\
\pm\frac{5}{1} & & \pm\frac{5}{2} & & \pm\frac{5}{3} & & \pm\frac{5}{4} & & \pm\frac{5}{5} & & \pm\frac{5}{6} & \cdots \\
 & \swarrow & & & & & & & & & & \\
\pm\frac{6}{1} & & \pm\frac{6}{2} & & \pm\frac{6}{3} & & \pm\frac{6}{4} & & \pm\frac{6}{5} & & \pm\frac{6}{6} & \cdots \\
\downarrow & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots &
\end{array}
$$

- Define the shifted sets $V_k := V + q_k := \{v + q_k : v \in V\}$.
- Then $[0,1] \subseteq \biguplus_{k=1}^{\infty} V_k \subseteq [-1,2]$, since:
  i) $\bigcup_{k=1}^{\infty} V_k \subseteq [-1,2]$, $k \in \mathbb{N}$. *Proof.* $V_k \subseteq [-1,2]$, $k \in \mathbb{N}$.
  ii) $[0,1] \subseteq \bigcup_{k=1}^{\infty} V_k$. *Proof.* Let $x \in [0,1]$. Since "$\sim$" partitions $[0,1]$ into its equivalence classes, we have $x \in [v]$ for some $v \in V \Rightarrow x - v = q_k$ for

some $k \Rightarrow x = v + q_k \in V_k$.

iii) The V_k's are pairwise disjoint.

Proof. If $x \in V_k \cap V_j$ for some $k \neq j$, then $x = v_k + q_k$ and $x = v_j + q_j$ for some $v_k, v_j \in V$ and distinct $q_k, q_j \in \mathbb{Q} \cap [-1, 1]$. Thus $v_k = x - q_k \underset{k \neq j}{\neq} x - q_j = v_j$, so that $\exists\, v_k, v_j \in V : v_k \neq v_j$ but $v_k - x = -q_k \in \mathbb{Q}$ and $v_j - x = -q_j \in \mathbb{Q}$, so $v_k \sim x$ and $v_j \sim x \Rightarrow v_k \sim v_j$ which contradicts the definition of V (V contains precisely one element of each equivalence class). $\square$

- $[0,1] \subseteq \biguplus_{k=1}^{\infty} V_k \subseteq [-1,2] \underset{\text{mon.}}{\Rightarrow} 1 \underset{1)}{=} \lambda([0,1]) \leq \lambda(\biguplus_{k=1}^{\infty} V_k) \leq \lambda([-1,2]) \underset{1)}{=} 3 \underset{3)}{\Rightarrow} 1 \leq \sum_{k=1}^{\infty} \lambda(V_k) \leq 3 \underset{2)}{\Rightarrow} 1 \leq \sum_{k=1}^{\infty} \lambda(V) \leq 3$ ↯ $\square$

The Vitali set V is known as a *non-measurable set*, a set we cannot reasonably measure (assign a volume to). In $d > 1$, we can consider $V \times [0,1]^{d-1}$ as a non-measurable set.

Question: How about weakening σ-additivity to finitely-many sets only?

Theorem 2.2 (Banach and Tarski (1924))
Let $d \geq 3$, $A, B \subseteq \mathbb{R}^d$ bounded, non-empty interior. Then $\exists\, k \in \mathbb{N}$ and partitions $A = \biguplus_{i=1}^k A_i$, $B = \biguplus_{i=1}^k B_i$ such that A_i, B_i are congruent $\forall\, i = 1, \ldots, k$.

- *Banach–Tarski paradox.* More colloquial, a pea can be chopped up and reassembled into the sun ($\Rightarrow$ buy gold, double it). But the partition elements are not easily constructed, their volumes are impossible to define (since $\lambda(A_i) = \lambda(B_i)$ for all i, at least one of the sets must be non-measurable, otherwise $A = B$).
- For countable Ω, one can always define λ or more general measures μ on $\mathcal{P}(\Omega)$ (see later), but for uncountable Ω, $\mathcal{P}(\Omega)$ can contain non-measurable sets (e.g. Vitali sets). $\mathcal{P}(\Omega)$ is thus too large to be useful for measuring volumes.
- Instead, we need to define λ or more general μ on a family of sets $\mathcal{F} \subsetneq \mathcal{P}(\Omega)$ that is *closed* w.r.t. certain set operations (i.e. performing these operations on sets in $\mathcal{F}$ yields a set in $\mathcal{F}$).

Question: The construction being put aside for now, what are the types of sets $\mathcal{F}$ we can construct measures μ on (and what are their properties)?

2.2 Systems of sets

In the construction of measures, several *systems of sets* (families of sets satisfying certain properties) play a role, including the following (later also π-systems and Dynkin systems).

Definition 2.3 (Semiring)
$\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a *semiring* on Ω if

i) $\emptyset \in \mathcal{A}$;

ii) $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}$; and

iii) $A, B \in \mathcal{A} \Rightarrow A \backslash B = \biguplus_{i=1}^{n} A_i$ for some $n \in \mathbb{N}$ and $A_1, \ldots, A_n \in \mathcal{A}$ with $A_i \cap A_j = \emptyset \; \forall \, i \neq j$.

Definition 2.4 (Ring)
$\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a *ring* on Ω if

i) $\emptyset \in \mathcal{A}$;

ii) $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$; and

iii) $A, B \in \mathcal{A} \Rightarrow A \backslash B \in \mathcal{A}$.

Definition 2.5 (Algebra)
$\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an *algebra* (or *field*) on Ω if

i) $\Omega \in \mathcal{A}$;

ii) $A \in \mathcal{A} \Rightarrow A^c = \Omega \backslash A \in \mathcal{A}$; and

iii) $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$.

Definition 2.6 (σ-algebra)
$\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an *σ-algebra* (or *σ-field*) on Ω if
i) $\Omega \in \mathcal{A}$;
ii) $A \in \mathcal{A} \Rightarrow A^c = \Omega \backslash A \in \mathcal{A}$; and
iii) $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Proposition 2.7 (σ-algebra $\subsetneq$ algebra $\subsetneq$ ring $\subsetneq$ semiring)
Every σ-algebra is an algebra, every algebra a ring, and every ring a semiring on Ω, with the inclusions being strict. An algebra on a finite set Ω is a σ-algebra. If Ω is an element of a ring, the ring is an algebra. If a semiring is closed wrt the union of two sets, the semiring is a ring.

Proof.

- σ-algebra $\subsetneq$ algebra: Take $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{P}(\Omega)$ with $A_n = \emptyset \; \forall\, n \geq 3 \Rightarrow$ D. 2.5 iii). **Strictness:** Consider $\Omega = (0,1]$, $\mathcal{A} = \{\biguplus_{i=1}^n (a_i, b_i] : 0 \leq a_i \leq b_i \leq 1$ for some $n \in \mathbb{N}\}$ is an algebra but since $(0,1) = \bigcup_{n=1}^\infty (0, 1-1/n] \notin \mathcal{A}$, $\mathcal{A}$ is not a σ-algebra. If $|\Omega| < \infty$, every countable union of sets in $\mathcal{A}$ is a finite union, so an algebra on Ω is also a σ-algebra.
- algebra $\subsetneq$ ring: $\emptyset = \Omega^c \underset{\text{i),ii)}}{\in} \mathcal{A}$. This implies that if $A, B \in \mathcal{A}$, then $A \backslash B = A \cap B^c \underset{\text{De Morgan}}{=} (A^c \cup B)^c \underset{\text{ii),iii)}}{\in} \mathcal{A} \Rightarrow$ D. 2.4 iii). **Strictness:** On $\Omega = \{1,2,3,4\}$, $\mathcal{A} = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$ is a ring (check) but not an algebra since $\{1,2,3\}^c = \{4\} \notin \mathcal{A}$. If $\Omega \in \mathcal{A}$, then $A^c = \Omega \backslash A \in \mathcal{A}$, so the ring $\mathcal{A}$ is also an algebra.
- ring $\subsetneq$ semiring: $A \cap B \underset{\text{De Morgan}}{=} A \cap (A \cap B^c)^c = A \backslash (A \backslash B) \underset{A\backslash B \,\in\, \mathcal{A}}{\in} \mathcal{A} \Rightarrow$ D. 2.3 ii). And $A, B \in \mathcal{A} \underset{\mathcal{A} \text{ ring}}{\Rightarrow} A \backslash B \in \mathcal{A}$ and thus $A \backslash B$ satisfies D. 2.3 iii). **Strictness:** On $\Omega = \{1,2,3,4\}$, $\mathcal{A} = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2,3\}\}$ is a semiring (check) but not a ring since $\{1\} \cup \{2\} = \{1,2\} \notin \mathcal{A}$. If $\mathcal{A}$ is closed wrt the union of two sets, then by induction also for finitely many, so also for finitely many pairwise disjoint sets $\Rightarrow A \backslash B = \biguplus_{i=1}^n A_i \in \mathcal{A}$. $\square$

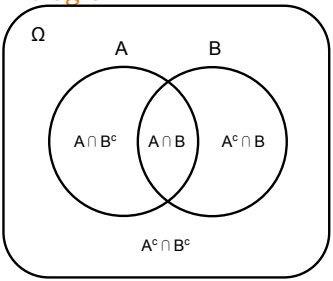
Remark 2.8 (About σ-algebras)

- A σ-algebra $\mathcal{F}$ (typical notation) is a family of subsets of Ω that includes Ω, is closed under complements and countable unions.
- σ-algebras are also closed w.r.t. countable intersections since $\bigcap_{i=1}^{\infty} A_i \underset{\text{De Morgan}}{=} (\bigcup_{i=1}^{\infty} A_i^c)^c \in \mathcal{F}$. So σ-algebras contain complements, countable unions and countable intersections, complements of such sets, etc. Apart from special cases, it seems hopeless to imagine all sets in $\mathcal{F}$.

Example 2.9 (Examples of σ-algebras)

1) The *trivial σ-algebra* $\mathcal{F} = \{\emptyset, \Omega\}$ is the smallest σ-algebra (contained in every σ-algebra) and the power set $\mathcal{F} = \mathcal{P}(\Omega)$ is the largest σ-algebra.
2) Let $A \subseteq \Omega$. Then $\mathcal{F} = \{\emptyset, A, A^c, \Omega\}$ is a σ-algebra.
3) Let $A, B \subseteq \Omega$, $A \nsubseteq B$, $B \nsubseteq A$, $A \cap B \neq \emptyset$, $A \cup B \neq \Omega$. Then $\mathcal{F} = \{\emptyset, A \cap B, A \cap B^c, A^c \cap B, A^c \cap B^c, A, A^c, B, B^c, (A \cap B) \cup (A^c \cap B^c), (A \cap B^c) \cup (A^c \cap B), (A \cap B)^c, (A \cap B^c)^c, (A^c \cap B)^c, (A^c \cap B^c)^c, \Omega\}$ is a σ-algebra.
   Construction: The four disjoint intersections $A \cap B, A \cap B^c, A^c \cap B, A^c \cap B^c$ partition Ω. Imagine taking the union of 0 of these 4 elements to form a new

set (so Ω), then of 1 of these 4 (so each at a time), then of 2 (all unions of two of these elements; only 2 not covered yet), then of 3 (so the complement of each), then of all 4 (so Ω); clearly, there are $2^4 = 16$ sets. Such ideas are best imagined with a *Venn diagram*:



With 3 sets, this can be up to $2^8 = 256$ sets already.

4) Let Ω be any set. The *countable-cocountable σ-algebra* $\mathcal{F} = \{A \subseteq \Omega : A \text{ is countable or } A^c \text{ is countable}\}$ is a σ-algebra:

   i) $\Omega^c = \emptyset$ is countable $\Rightarrow \Omega \in \mathcal{F}$;

   ii) $A \in \mathcal{F} \Rightarrow A$ is countable or A^c is countable $\Rightarrow (A^c)^c$ is countable or A^c is countable $\Rightarrow A^c$ is countable or $(A^c)^c$ is countable $\Rightarrow A^c \in \mathcal{F}$.

iii) $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{F}$.

Case 1: A_i countable $\forall\, i \in \mathbb{N} \underset{\text{diag. arg.}}{\overset{\text{Cantor's first}}{\Rightarrow}} \bigcup_{i=1}^{\infty} A_i$ countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Case 2: $\exists\, k \in \mathbb{N} : A_k$ uncountable $\underset{A_k \,\in\, \mathcal{F}}{\Rightarrow} A_k^c$ countable $\Rightarrow (\bigcup_{i=1}^{\infty} A_i)^c \underset{\text{De Morgan}}{=} \bigcap_{i=1}^{\infty} A_i^c \subseteq A_k^c$ countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

5) Let $\mathcal{F}$ be a σ-algebra on Ω and $\Omega' \subseteq \Omega$. The *trace σ-algebra*

$$\mathcal{F}' = \mathcal{F}\big|_{\Omega'} := \{A \cap \Omega' : A \in \mathcal{F}\}$$

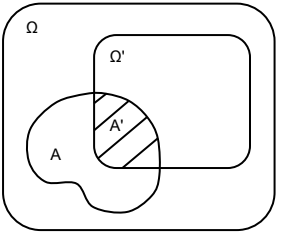
of Ω' in $\mathcal{F}$ is a σ-algebra on Ω':

i) $\Omega' = \Omega \cap \Omega' \underset{\Omega \,\in\, \mathcal{F}}{\in} \mathcal{F}'$;

ii) $A' \in \mathcal{F}' \Rightarrow \exists\, A \in \mathcal{F}$ such that $A' = A \cap \Omega' \Rightarrow (A')^{c_{\Omega'}} = (A \cap \Omega')^{c_{\Omega'}} \underset{\text{De Morgan}}{=} A^{c_{\Omega'}} \cup (\Omega')^{c_{\Omega'}} = A^{c_{\Omega'}} \cup \emptyset = A^{c_{\Omega'}} = A^{c_\Omega} \cap \Omega' \underset{A^{c_\Omega} \,\in\, \mathcal{F}}{\in} \mathcal{F}'$;

iii) $\{A_i'\}_{i\in\mathbb{N}} \subseteq \mathcal{F}' \Rightarrow \forall\, i \in \mathbb{N}\ \exists\, A_i \in \mathcal{F} : A_i' = A_i \cap \Omega' \Rightarrow \bigcup_{i=1}^{\infty} A_i' = \bigcup_{i=1}^{\infty} (A_i \cap \Omega') \underset{\text{distr.}}{=} \left(\bigcup_{i=1}^{\infty} A_i\right) \cap \Omega' \underset{\bigcup_{i=1}^{\infty} A_i \,\in\, \mathcal{F}}{\in} \mathcal{F}'$.

Venn diagram:



6) Let $X : \Omega \to \Omega'$ be a map between two sets Ω, Ω', and let $\mathcal{F}'$ be a σ-algebra on Ω'. Then

$$\mathcal{F} = \sigma(X) := X^{-1}(\mathcal{F}') = \{X^{-1}(A') : A' \in \mathcal{F}'\}$$

is a σ-algebra on Ω, the *preimage σ-algebra* or *σ-algebra generated by X*:

i) $\Omega = X^{-1}(\Omega') \underset{\Omega' \in \mathcal{F}'}{\in} \mathcal{F}$;

ii) $A \in \mathcal{F} \underset{\text{def.}}{\Rightarrow} \exists\, A' \in \mathcal{F}' : A = X^{-1}(A') \Rightarrow A^{c_\Omega} = (X^{-1}(A'))^{c_\Omega} \underset{\text{S. 1.2}}{=} X^{-1}(A'^{c_{\Omega'}}) \underset{A'^{c_{\Omega'}}}{\in} \mathcal{F}$;

iii) $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{F} \Rightarrow \forall\, i \in \mathbb{N},\ A_i = X^{-1}(A_i')$ for some $A_i' \in \mathcal{F}' \Rightarrow \bigcup_{i=1}^\infty A_i = \bigcup_{i=1}^\infty X^{-1}(A_i') \underset{\text{S. 1.2}}{=} X^{-1}(\bigcup_{i=1}^\infty A_i') \underset{\bigcup_{i=1}^\infty A_i' \in \mathcal{F}'}{\in} \mathcal{F}$.

7) A *filtration* is an increasing sequence $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ of σ-algebras and can be used to model *information accrual* over time.

Example: Consider modeling infinite coin tosses with

$$\Omega = \{0,1\}^\infty = \{\boldsymbol{\omega} = (\omega_1, \omega_2, \dots) : \omega_i \in \{0,1\}\ \forall\, i \in \mathbb{N}\}.$$

Let $\mathcal{F}_n = \Big\{\{\boldsymbol{\omega} \in \Omega : (\omega_1, \dots, \omega_n) \in A\}$ for some $A \subseteq \{0,1\}^n\Big\}$ model all events whose occurence can be decided after the first n tosses (e.g. $B = \{\boldsymbol{\omega} \in \Omega : \omega_3 = 1\} \in \mathcal{F}_3$ but $B \notin \mathcal{F}_2$).

Then:

- $\forall\, n \in \mathbb{N}$, $\mathcal{F}_n = \Big\{\{\boldsymbol{\omega} \in \Omega : (\omega_1, \ldots, \omega_n) \in A\}$ for some $A \subseteq \{0,1\}^n\Big\}$ is a σ-algebra on Ω:
  i) $A = \{0,1\}^n \Rightarrow \Omega \in \mathcal{F}_n$;
  ii) If $B \in \mathcal{F}_n$ for some $A \subseteq \{0,1\}^n$, then B^c is obtained for $A^c \subseteq \{0,1\}^n$ $\Rightarrow B^c \in \mathcal{F}_n$;
  iii) If $\{B_i\}_{i\in\mathbb{N}} \subseteq \mathcal{F}_n$ with corresponding sets $\{A_i\}_{i\in\mathbb{N}} \subseteq \{0,1\}^n \Rightarrow \bigcup_{i=1}^\infty B_i = \{\{\boldsymbol{\omega} \in \Omega : (\omega_1, \ldots, \omega_n) \in A\}$ for $A = \bigcup_{i=1}^\infty A_i\} \underset{\bigcup_{i=1}^\infty A_i \subseteq \{0,1\}^n}{\in} \mathcal{F}_n$.

- $\mathcal{F} := \bigcup_{i=1}^\infty \mathcal{F}_i$ is an algebra but not a σ-algebra:
  i) $\mathcal{F}_1$ σ-algebra $\Rightarrow \Omega \in \mathcal{F}_1 \underset{\mathcal{F}_1 \subseteq \mathcal{F}}{\Rightarrow} \Omega \in \mathcal{F}$;
  ii) $A \in \mathcal{F} \Rightarrow \exists\, i \in \mathbb{N} : A \in \mathcal{F}_i \underset{\mathcal{F}_i\ \sigma\text{-algebra}}{\Rightarrow} A^c \in \mathcal{F}_i \Rightarrow A^c \in \mathcal{F}$;
  iii) For $A_1, \ldots, A_n \in \mathcal{F}$, $\exists\, j_1, \ldots, j_n \in \mathbb{N} : A_i \in \mathcal{F}_{j_i}$, $i = 1, \ldots, n \Rightarrow$ $A_1, \ldots, A_n \in \mathcal{F}_{\max\{j_1,\ldots,j_n\}} \underset{\mathcal{F}_{\max}\ \sigma\text{-algebra}}{\Rightarrow} \bigcup_{i=1}^n A_i \in \mathcal{F}_{\max\{j_1,\ldots,j_n\}} \subseteq \mathcal{F}$.

  However, let $A_i = \{\boldsymbol{\omega} \in \Omega : \omega_i = 1\}$, $i \in \mathbb{N}$, and $A_{2\mathbb{N}} = \{\boldsymbol{\omega} \in \Omega : \omega_{2\mathbb{N}} = 1\}$. Then $A_i \in \mathcal{F}_i \subseteq \mathcal{F}\ \forall\, i \in \mathbb{N}$, but $A_{2\mathbb{N}} = \bigcap_{i\in 2\mathbb{N}}^\infty A_i \notin \mathcal{F}$ (since the occurrence of $A_{2\mathbb{N}}$ cannot be determined in n tosses for any finite n).